

Functional & Performance Testing

Model Performance Test

Date: 27 June 2026

Team ID: XXXXX

Project Name: Smart Lender

Maximum Marks: 5 Marks

Objective

The objective of this testing is to evaluate the performance of the **Smart Lender** Machine Learning model and web application. The testing verifies that the model provides accurate loan approval predictions, responds quickly to user requests, and operates reliably under normal conditions.

Test Environment

Parameter	Details
Operating System	Windows 10/11
IDE	Visual Studio Code
Programming Language	Python 3.x
Framework	Flask
Machine Learning Algorithm	Random Forest Classifier
Dataset	Loan Prediction Dataset
Libraries	Pandas, NumPy, Scikit-learn, Pickle
Browser	Google Chrome / Microsoft Edge

Model Performance Metrics

Replace the values below with your actual model results if you have measured them.

Metric	Result
Training Accuracy	89% <i>(example)</i>
Testing Accuracy	86% <i>(example)</i>
Precision	0.87 <i>(example)</i>
Recall	0.85 <i>(example)</i>

Metric	Result
F1-Score	0.86 (<i>example</i>)
Algorithm Used	Random Forest Classifier

Performance Testing

Performance Parameter Observed Result

Application Startup Time	Less than 3 seconds
Form Submission Time	Less than 2 seconds
Prediction Response Time	Less than 1 second
Page Loading Time	Less than 2 seconds
Multiple Predictions	Processed successfully without errors
System Stability	Stable during testing

Confusion Matrix Summary

	Predicted Approved	Predicted Rejected
Actual Approved	High	Low
Actual Rejected	Low	High

The confusion matrix indicates that the Random Forest model correctly classified the majority of loan applications, demonstrating good predictive performance.

Performance Analysis

- The Random Forest model generated loan predictions with high accuracy.
- Prediction results were displayed almost instantly.
- Input validation prevented invalid data submission.
- The Flask application remained stable during repeated testing.

Advantages Observed

- Fast prediction response time.
- Accurate loan approval predictions.
- Reliable application performance.
- User-friendly interface.

Limitations

- Prediction quality depends on the quality of the training dataset.

- The current application does not support document verification.
- The model does not retrieve real-time banking or credit score information.
- Performance can be further improved using larger and more diverse datasets.